

5ª QUESTÃO

Valor: 1,0

Sejam a, b e c números reais positivos. Prove que

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9.$$

$$\frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$A(x) \geq H(x) \quad \square$$

$$a+b+c \geq 3\sqrt[3]{abc} \quad (1)$$

$$G(x) = \sqrt[n]{x_1 x_2 \dots x_n}$$

$$a \rightarrow \frac{1}{a} \quad , \quad b \rightarrow \frac{1}{b} \quad , \quad c \rightarrow \frac{1}{c}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3\sqrt[3]{\frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}} \quad (2)$$

$$(1) \times (2) : (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 3\sqrt[3]{abc} \times 3\sqrt[3]{\frac{1}{abc}}$$

$$(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9 \quad \square$$

$$A(\vec{x}) = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i \quad , \quad \vec{x} = \{x_i\}_{i=1}^n \quad , \quad x_i \in \mathbb{R}$$

$$H(x) = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} \quad , \quad \vec{x} = \{x_i\}_{i=1}^n \quad , \quad x_i \in \mathbb{R}^+$$

$$G(x) = \sqrt[n]{x_1 x_2 \dots x_n} = \left(\prod_{i=1}^n x_i \right)^{1/n} \quad ,$$

$$Q(x) = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}} = \frac{1}{\sqrt{n}} \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

$$\boxed{Q(x) \geq A(x) \geq G(x) \geq H(x)} \quad (*)$$

I) $Q(x) \geq A(x)$, $\boxed{\text{CAUCHY-SCHWARZ}} \quad (C.S)$
 $\vec{x}, \vec{y} \in \mathbb{R}^n$, $\vec{x} = \{x_i\}_{i=1}^n$, $\vec{y} = \{y_i\}_{i=1}^n$, $\forall x_i, y_i \in \mathbb{R}$

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right) \quad C-S$$

$$|\langle \vec{x}, \vec{y} \rangle|^2 = \left(\sum_{i=1}^n x_i y_i \right)^2$$

$$|\vec{x}|^2 = \sum_{i=1}^n x_i^2 \quad , \quad |\vec{y}|^2 = \sum_{i=1}^n y_i^2$$

$$|\langle \vec{x}, \vec{y} \rangle|^2 \leq |\vec{x}|^2 \cdot |\vec{y}|^2$$

$$y_i = 1, \quad \forall i \in [1, n]$$

$$\left(\sum_{i=1}^n x_i \cdot 1 \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \underbrace{\left(\sum_{i=1}^n 1 \right)}_{=n}$$

$$\therefore \left(\sum_{i=1}^n x_i \right)^2 \leq n \sum_{i=1}^n x_i^2$$

$$0 \leq \frac{1}{n^2} \left(\sum_{i=1}^n x_i \right)^2 \leq \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$\underbrace{\frac{1}{n} \sum_{i=1}^n x_i}_{=A(x)} \leq \underbrace{\sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}}_{=Q(x)} \quad \square$$

II) $A(x) \geq G(x)$ $\boxed{\text{JENSEN (J)}}$

$$x_1, x_2 \in \mathbb{R} \quad , \quad f[t x_1 + (1-t) x_2] \geq t f(x_1) + (1-t) f(x_2)$$

$$t \in [0, 1]$$

$$\ln a / \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \geq \frac{1}{n} \sum_{i=1}^n \ln a / x_i = \frac{1}{n} \ln a \sum_{i=1}^n \frac{1}{x_i}$$

$$\log \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \geq \frac{1}{n} \log x_1 + \frac{1}{n} \log x_2 + \dots + \frac{1}{n} \log x_n$$

$$\log \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \geq \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\boxed{\log(xy) = \log x + \log y}$$

$$\log \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \geq \frac{1}{n} \log \left(\prod_{i=1}^n x_i \right)$$

$$\left. \begin{aligned} \log(x_1 x_2 \dots x_n) &= \log x_1 + \log x_2 + \dots + \log x_n \\ \log \left(\prod_{i=1}^n x_i \right) &= \sum_{i=1}^n \log(x_i) \end{aligned} \right\}$$

$$\log \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \geq \log \left[\left(\prod_{i=1}^n x_i \right)^{1/n} \right] \leftarrow$$

$$\left. \begin{aligned} \frac{1}{b} \log a &= \log(a^{1/b}) \\ b \log a &= \log(a^b) \end{aligned} \right\}$$

$$\log \left(\frac{1}{n} \sum_{i=1}^n x_i \right) - \log \left[\left(\prod_{i=1}^n x_i \right)^{1/n} \right] \geq 0$$

$$\log \left[\underbrace{\frac{1}{n} \sum_{i=1}^n x_i - \left(\prod_{i=1}^n x_i \right)^{1/n}}_{\geq 0} \right] \geq 0$$

$$\frac{1}{n} \sum_{i=1}^n x_i - \left(\prod_{i=1}^n x_i \right)^{1/n} \geq 0$$

$$\underbrace{\frac{1}{n} \sum_{i=1}^n x_i}_{A(x)} - \underbrace{\left(\prod_{i=1}^n x_i \right)^{1/n}}_{G(x)} \geq 0$$

$$A(x) - G(x) \geq 0$$

$$\boxed{A(x) \geq G(x)} \quad \square$$

III) $G(x) \geq H(x)$

$$G(x) = \sqrt[n]{x_1 x_2 \dots x_n}$$

$$H(x) = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

$$A(x) = \frac{x_1 + x_2 + \dots + x_n}{n}, \quad A(1/x) = \frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n} = \frac{1}{H(x)}$$

$$\boxed{A(1/x) = \frac{1}{H(x)}} \Leftrightarrow H(1/x) = \frac{1}{A(x)}$$

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

$$x_i \rightarrow \frac{1}{x_i}$$

$$\frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n} \geq \sqrt[n]{\frac{1}{x_1} \cdot \frac{1}{x_2} \dots \frac{1}{x_n}}$$

$$\frac{1}{H(x)} \geq \frac{1}{\sqrt[n]{x_1 x_2 \dots x_n}} = \frac{1}{G(x)}$$

$$\frac{1}{H(x)} \geq \frac{1}{G(x)}$$

$$\boxed{G(x) \geq H(x)} \quad \square$$

I) ✓

II) ✓ $\Rightarrow Q(x) \geq A(x) \geq G(x) \geq H(x)$ $\square$

III) ✓

(I)

(II)

(III)